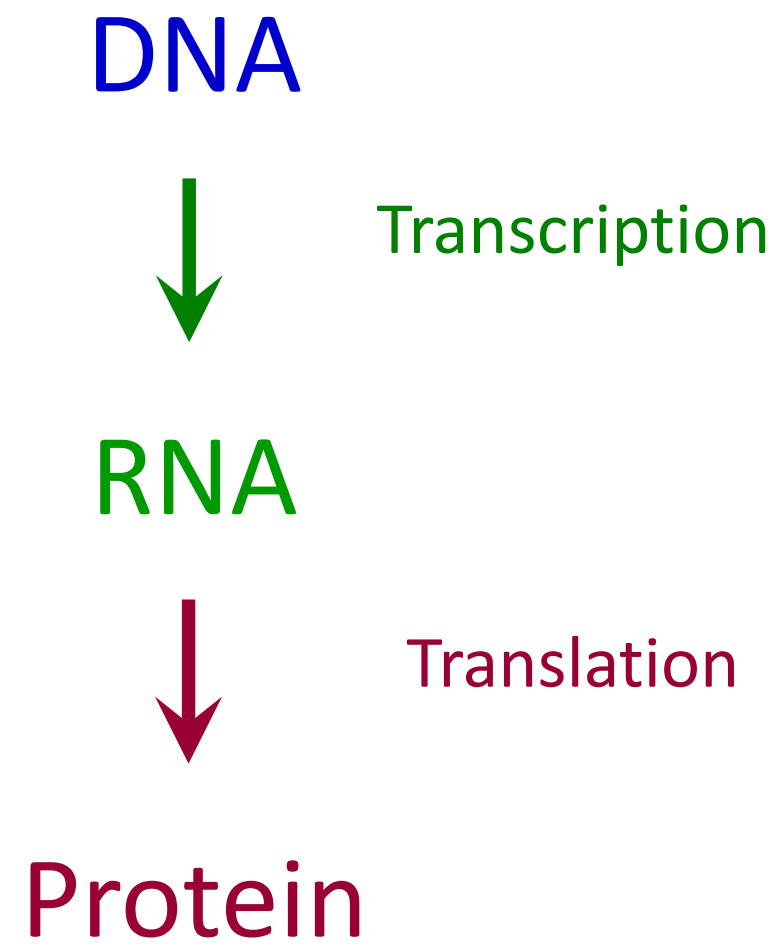
# Chapter III: Transcription, Translation


DNA
Transcription
RNA
Translation
Protein

# Definitions

Transcription – To copy down, within the same language

Language = Nucleic Acids

DNA to RNA

Translation – To translate from one language to another

From Nucleic Acids to Amino Acids

RNA to Protein

# DNA vs. RNA

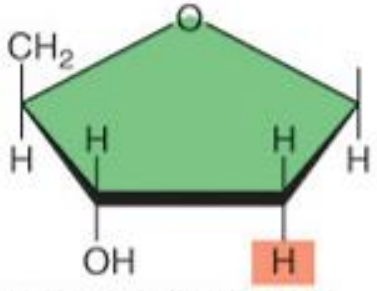


Double-stranded

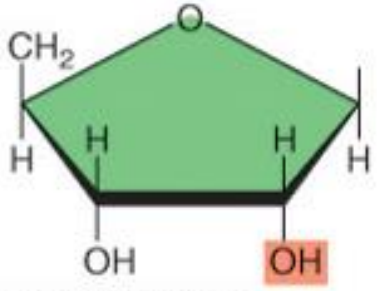


Generally single-stranded

b.



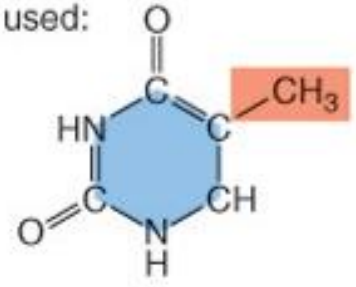

Deoxyribose as the sugar




Ribose as the sugar

c.

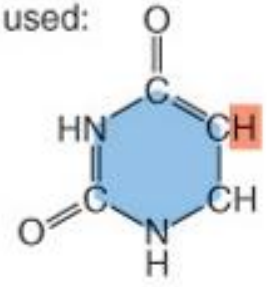
Bases used:




Thymine (T)
Cytosine (C)
Adenine (A)
Guanine (G)

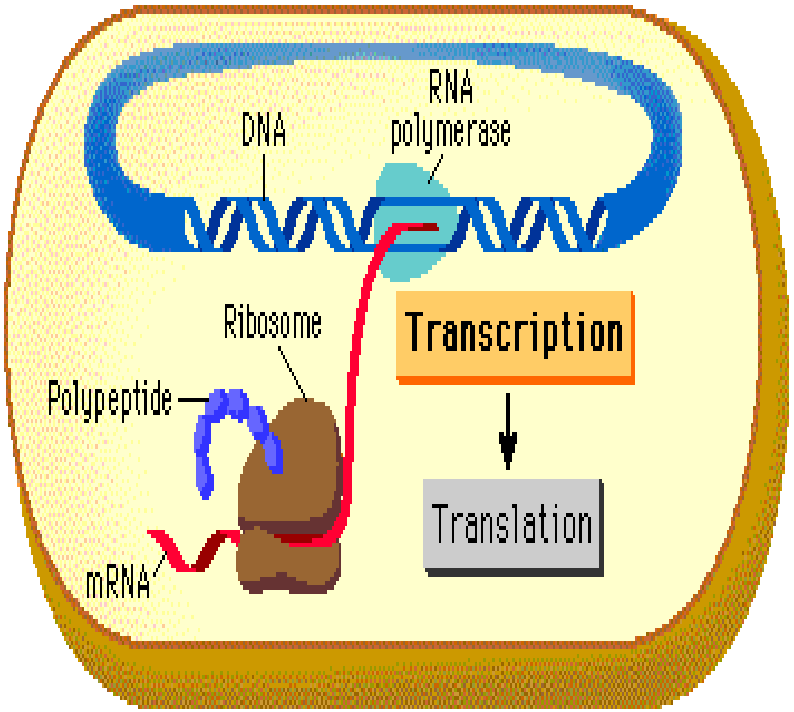
d.

Bases used:



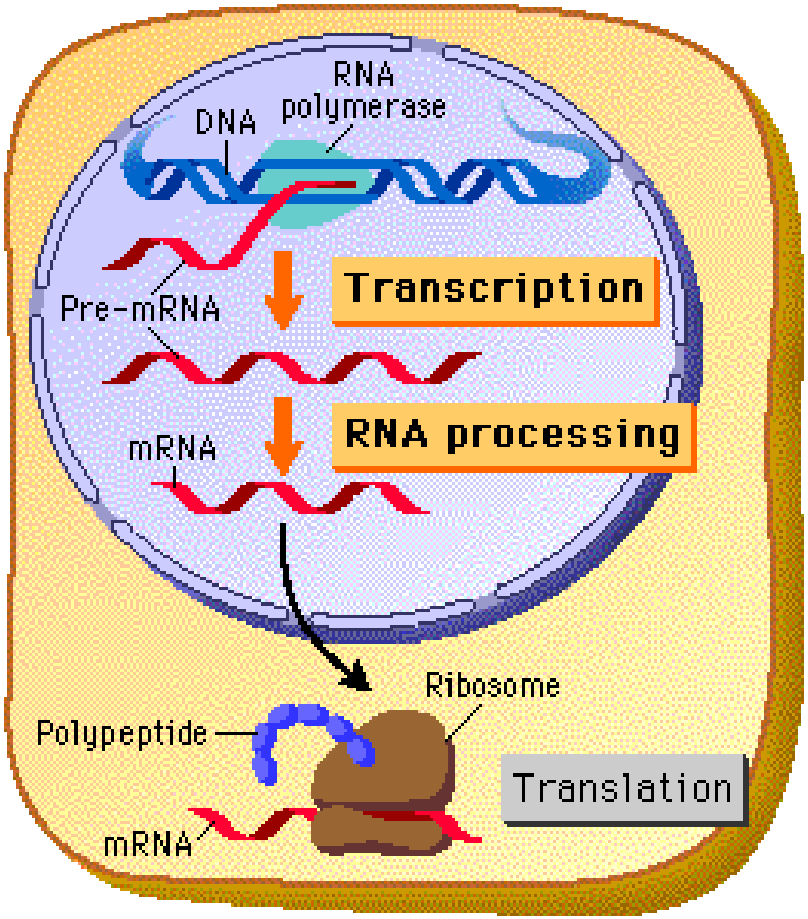

Uracil (U)
Cytosine (C)
Adenine (A)
Guanine (G)

# Prokaryotic transcription and translation




- It's a continuous process as both transcription and translation occur in cytoplasm.
- The ribosomes are 70S, each consisting of a small (30S) and a large (50S) subunit

# Eukaryotic transcription and translation



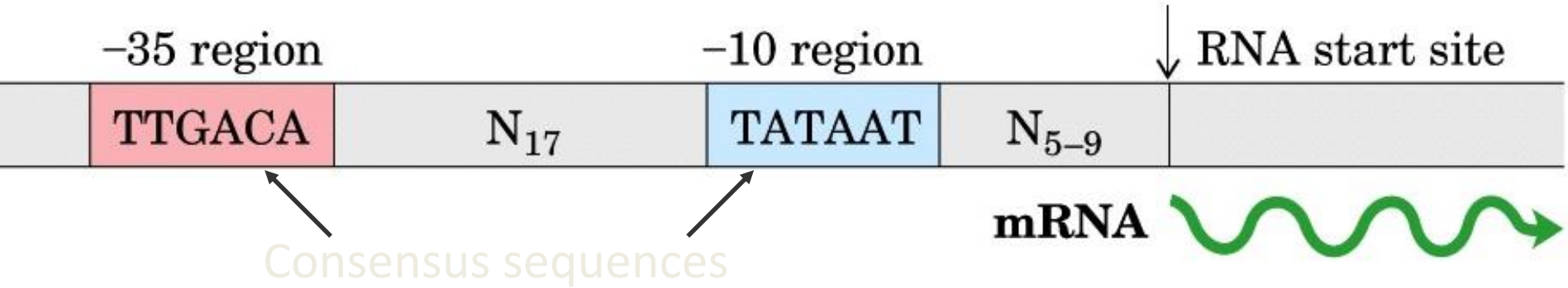

- Transcription occurs in **nucleus** and translation occur in **cytoplasm**.
- The ribosomes are 80S, each consisting of a small (40S) and a large (60S) subunit

# Transcription

# Transcription Recognition

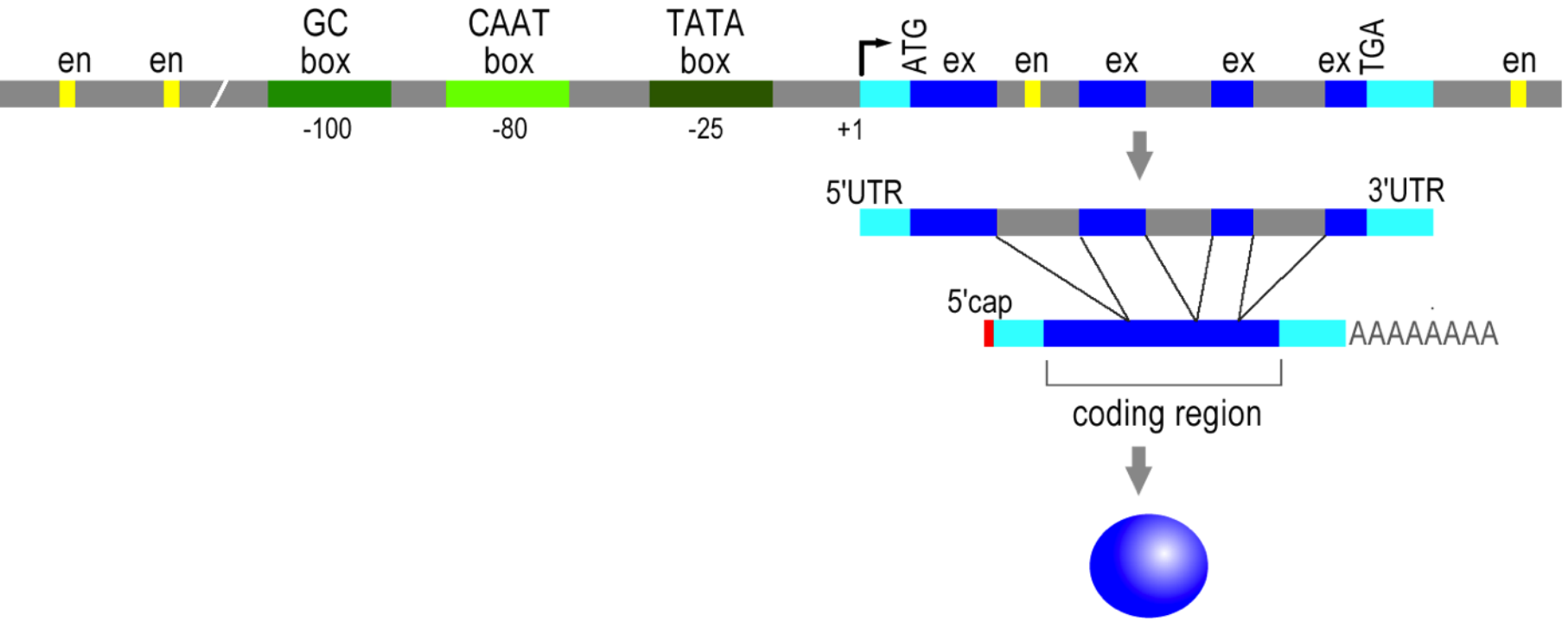
- Template strand
- Coding strand
- Promoters
  - Binding sites for RNA pol on template strand
  - ~40 bp of specific sequences with a specific order and distance between them.
- Core promoter elements for E. coli
  - -10 box (Pribnow box)
  - -35 box
- Numbers refer to distance from transcription start site

# Typical Prokaryote Promoter

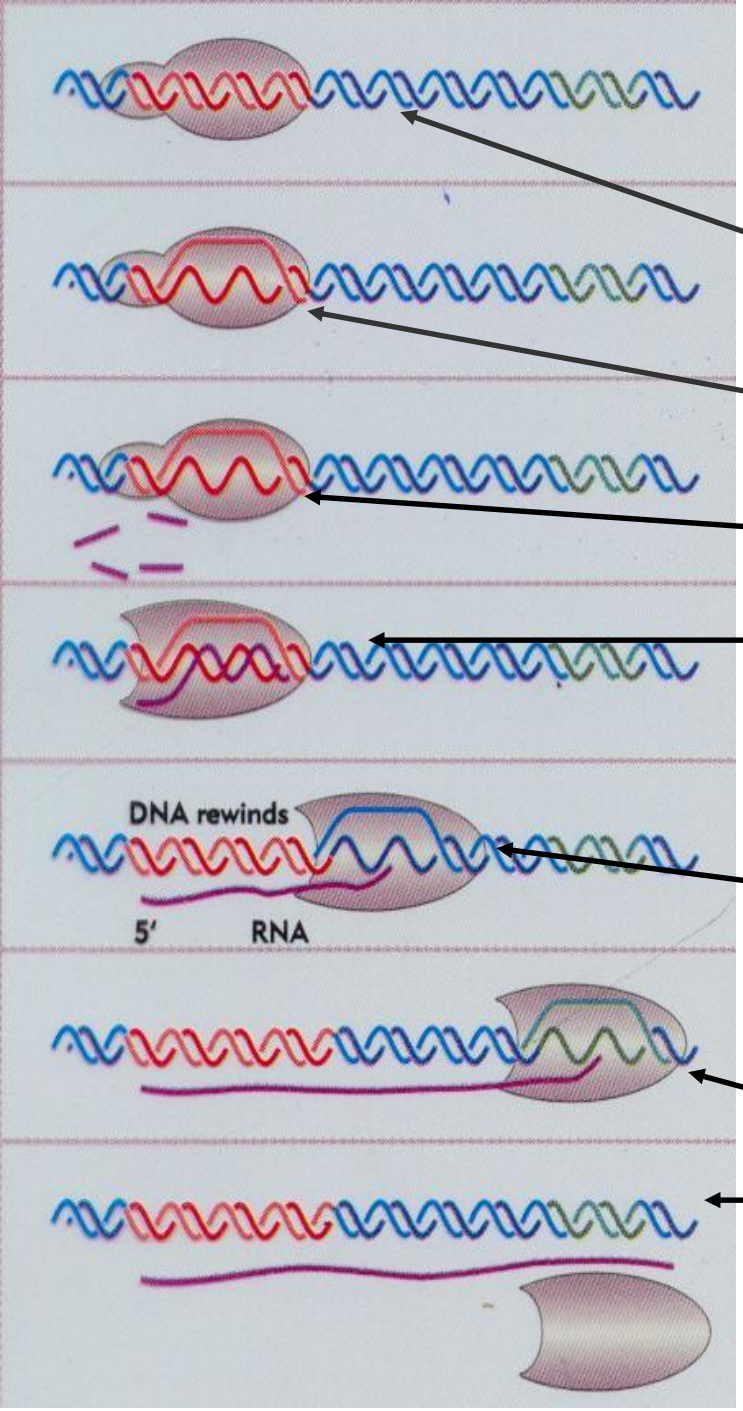



- Pribnow box located at –10 (6-7bp)
- -35 sequence ~(6bp)
- Consensus sequences: Strongest promoters match consensus
  - Up mutation: mutation that makes promoter more like consensus
  - Down Mutation: virtually any mutation that alters a match with the consensus

# Typical Eukaryot Promoter



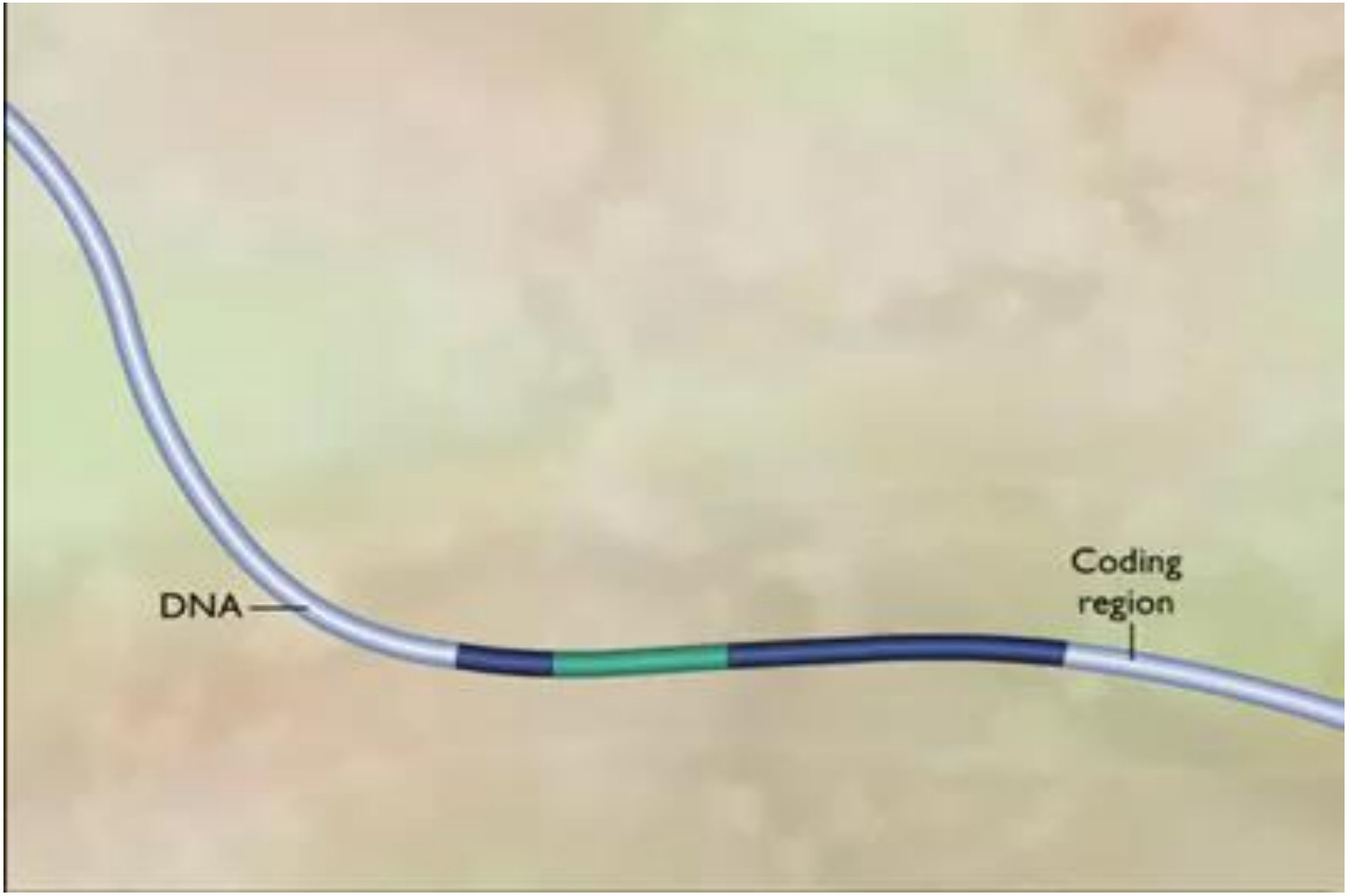

- **TATA box:** around -35 nu before the start transcription site.
- **CAAT box:** around -80 nu before the start transcription site.
- **GC island**: a region rich with GC content. Most of the DNA methylation focus on this region.



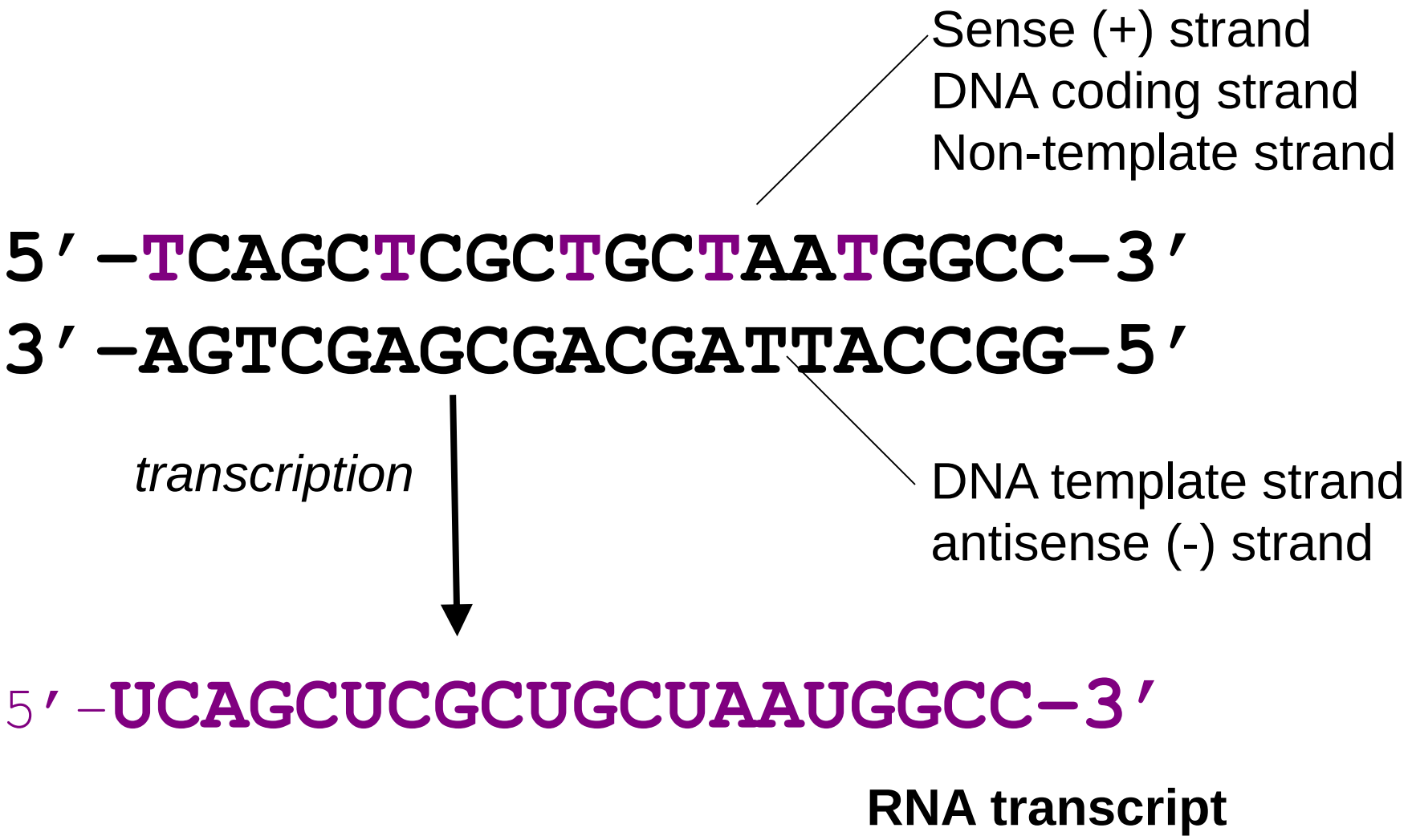

# Stages of Transcription

- Template recognition
  - RNA pol binds to DNA
  - DNA unwound
- Initiation
- Elongation
  - RNA pol moves and synthesizes RNA
  - Unwound region moves
- Termination
  - RNA pol reaches end
  - RNA pol and RNA released
  - DNA duplex reforms

# Watch clip



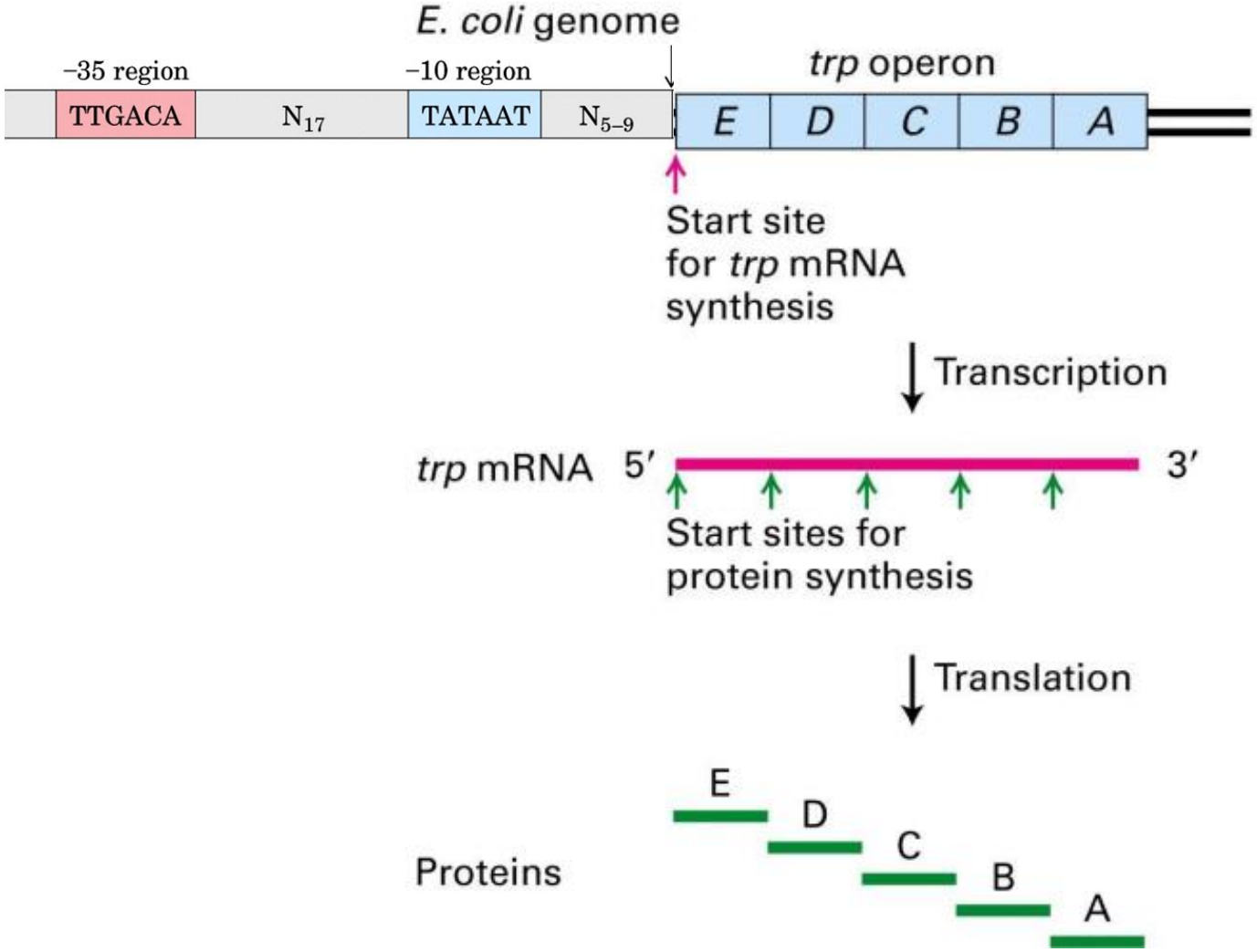
# Template and Coding Strands



# Differences between replication and transcription

| | replication | transcription |
|---|---|---|
| template | double strands | single strand |
| substrate | dNTP | NTP |
| primer | yes | no |
| Enzyme | DNA polymerase | RNA polymerase |
| product | dsDNA | ssRNA |
| base pair | A-T, G-C | A-U, T-A, G-C |

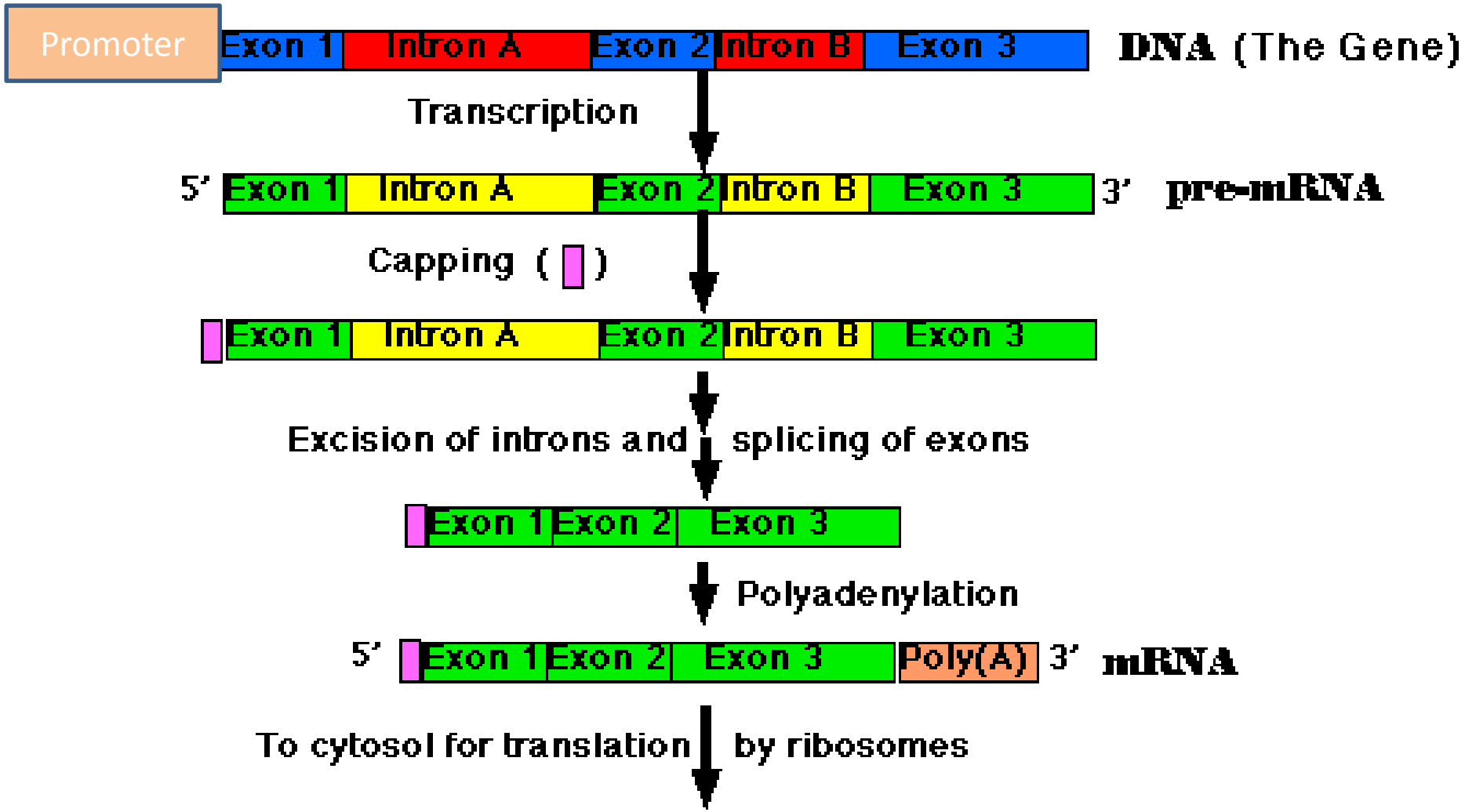
# Gene structure in prokaryotic cell



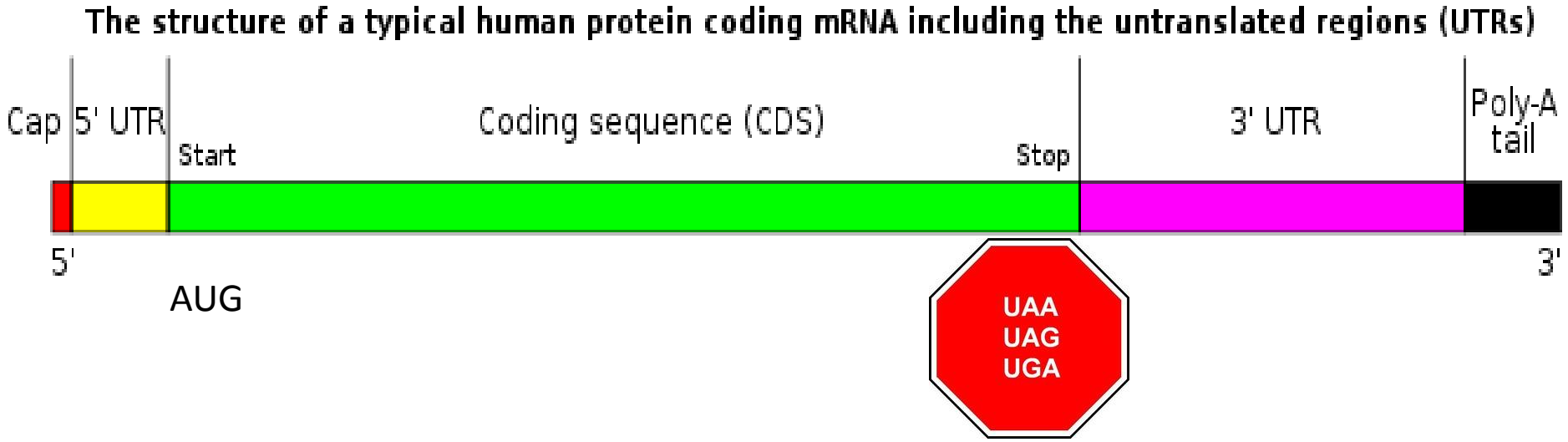
# Watch clip



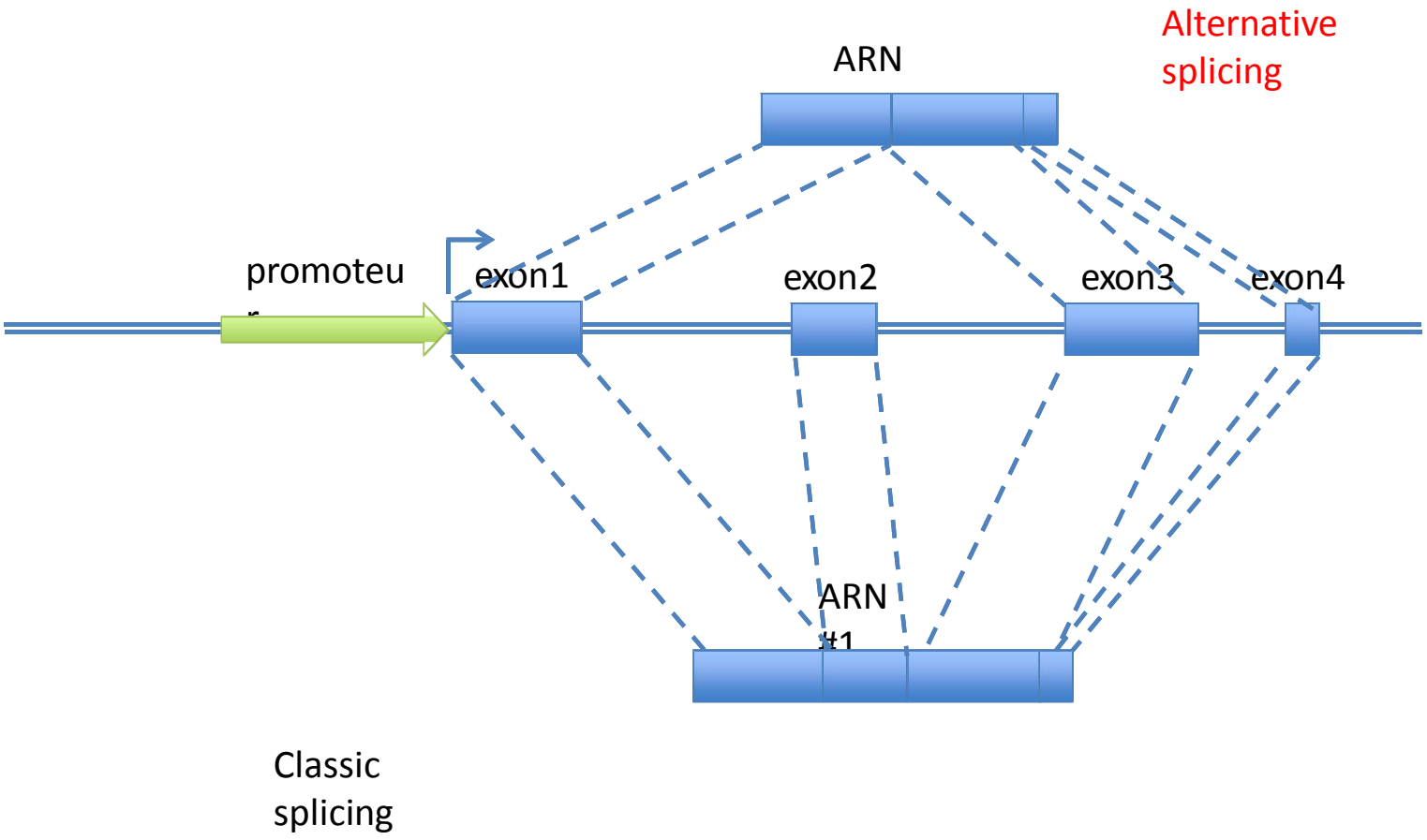
# Mature process of mRNA (only in eukaryotic cell)



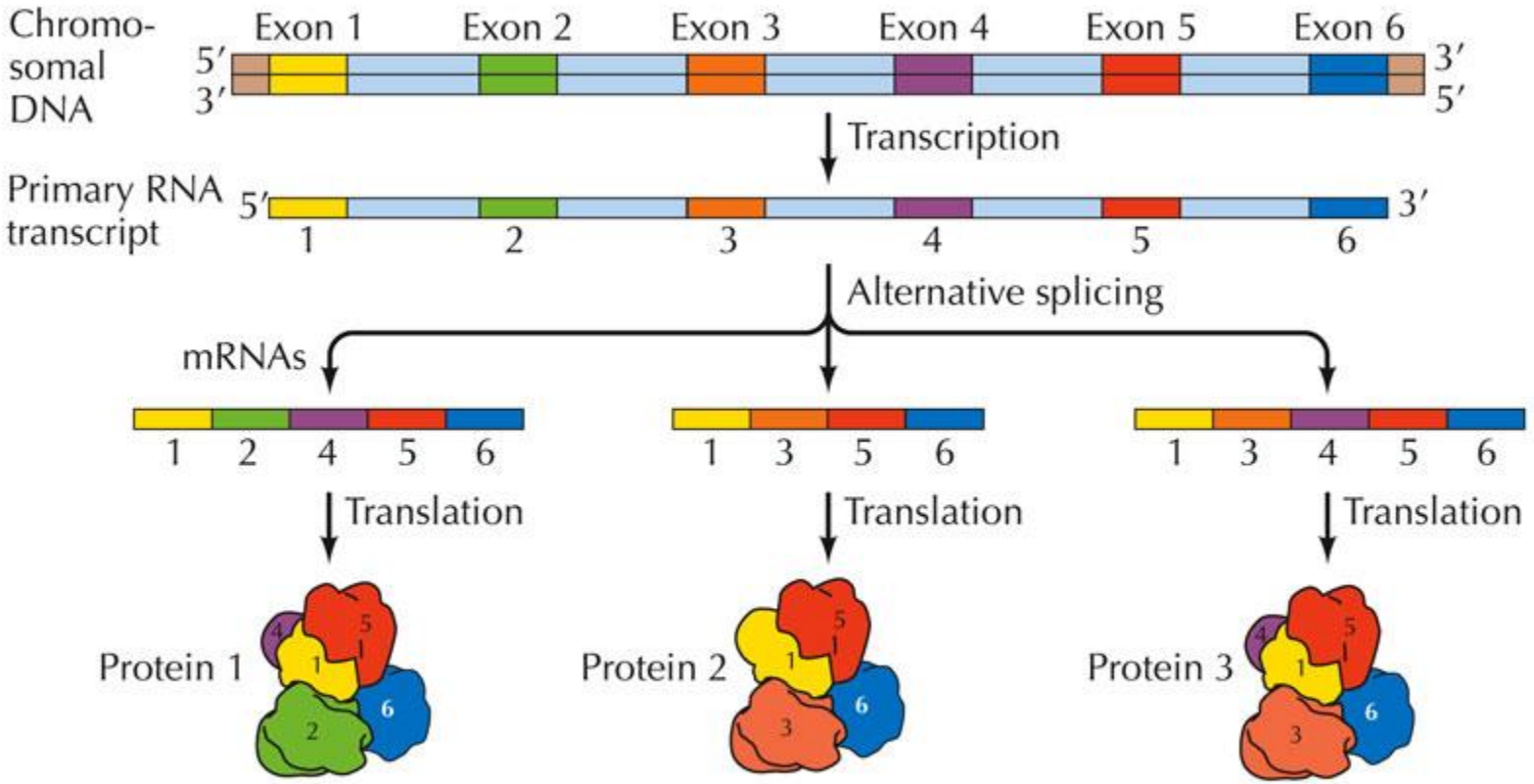
# Mature process of mRNA (only in eukaryotic cell)



# Alternative splicing: one gene can produce different mRNA



# Alternative splicing: one gene can produce different mRNA



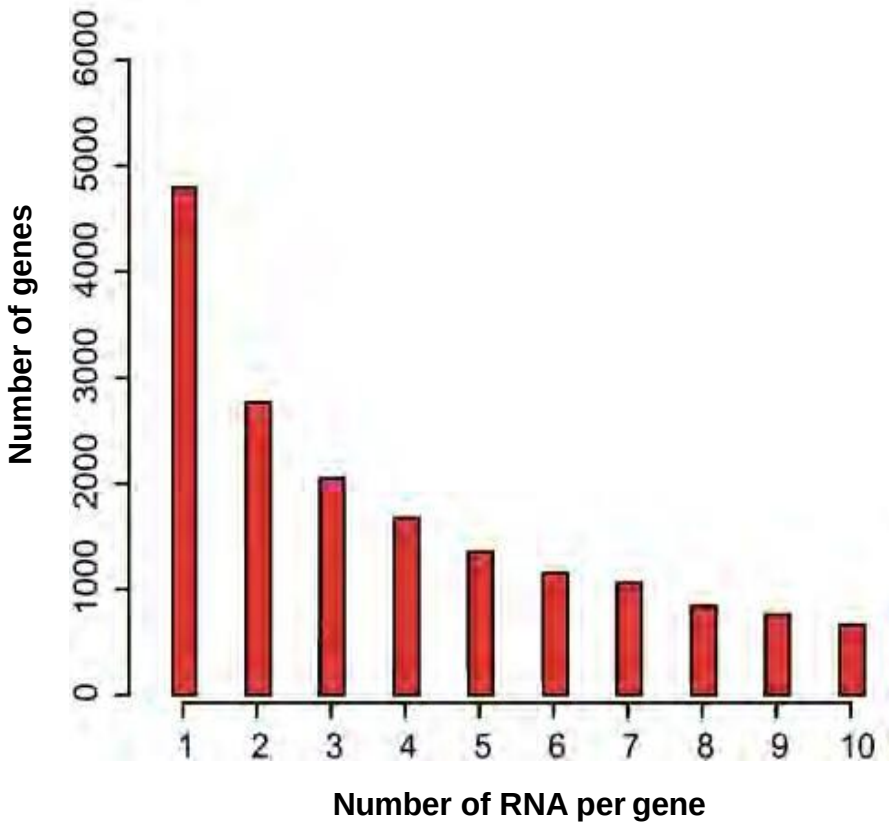

*THE CELL, Fourth Edition*, Figure 5.5 © 2006 ASM Press and Sinauer Associates, Inc.

# Alternative splicing is common in higher vertebrates

Mammals:

20000 coding genes

> 100 000 different mRNA

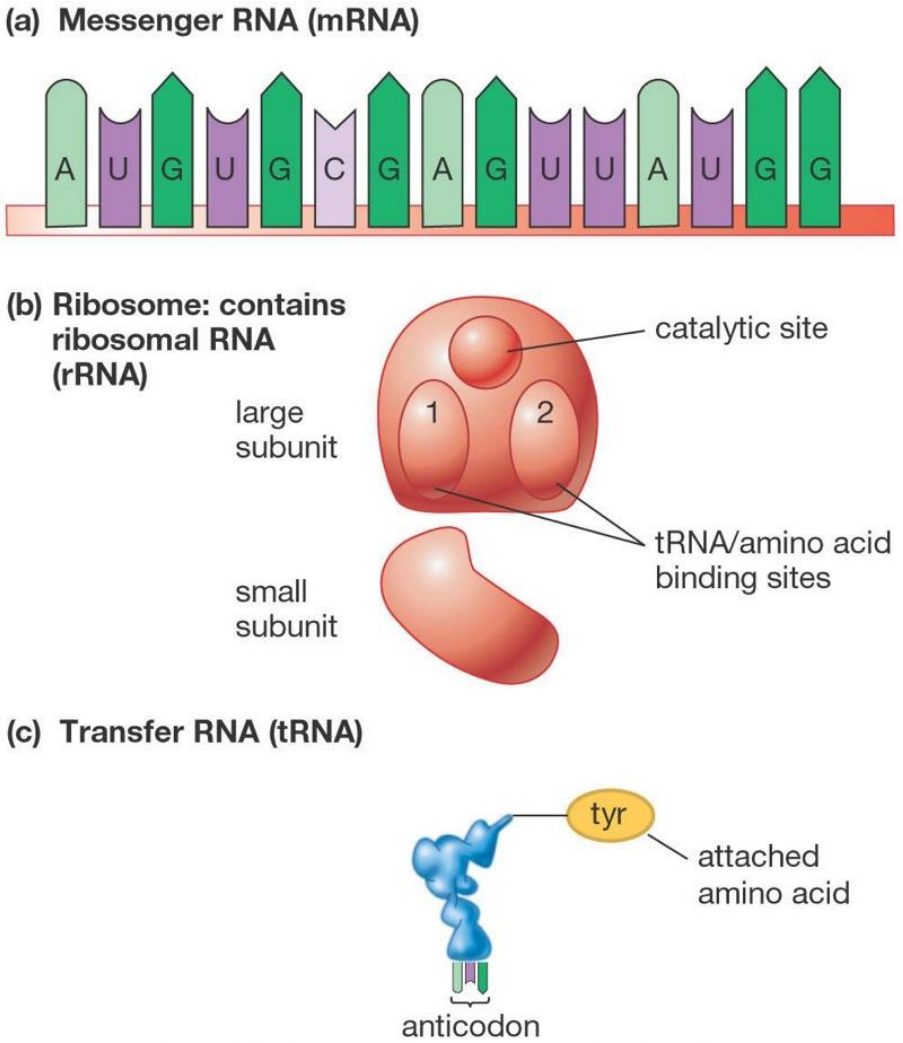



Derrien et al, Genome Res 22:1775.

# Translation

# Key Players

- mRNA carries the information from a gene in DNA.
- Ribosomes, made of rRNA, consist of subunits and carry out an enzyme-like role.
- tRNA carries specific amino acids to the ribosome.

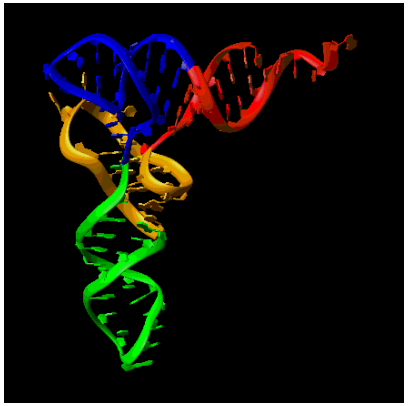



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# tRNA

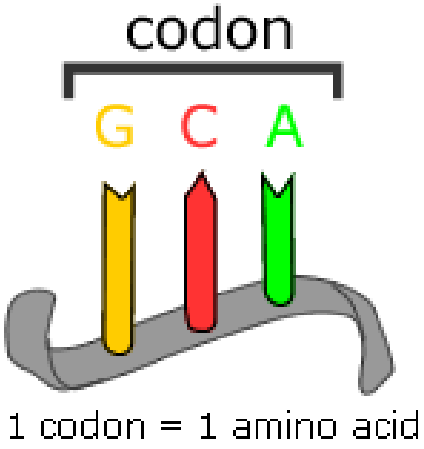



- Transfer RNA
- Bound to one amino acid on one end
- Anticodon on the other end complements mRNA codon

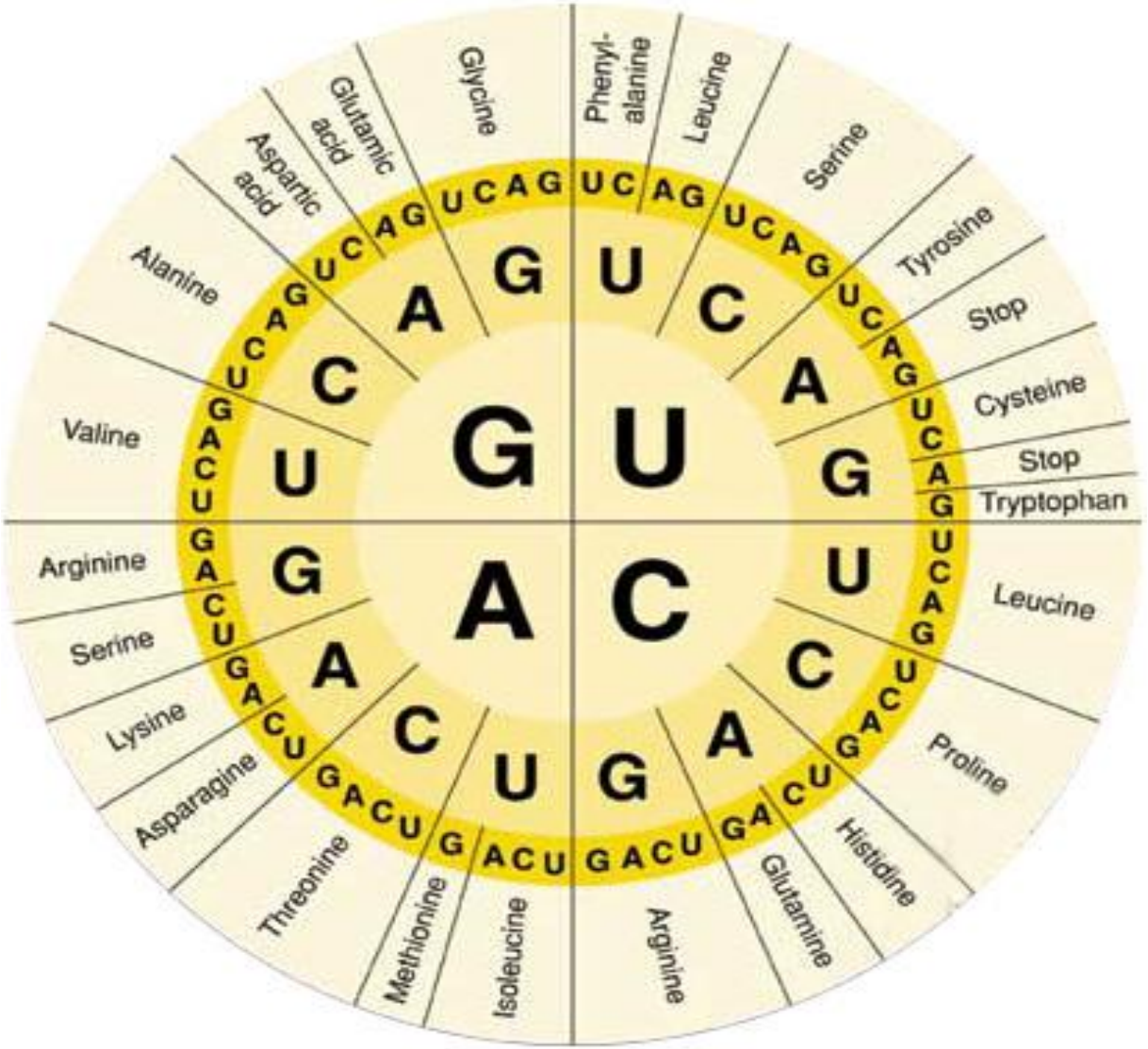


# Reading the DNA code

- Every 3 DNA bases pairs with 3 mRNA bases
- Every group of 3 mRNA bases encodes a single amino acid
- Codon- coding triplet of mRNA bases

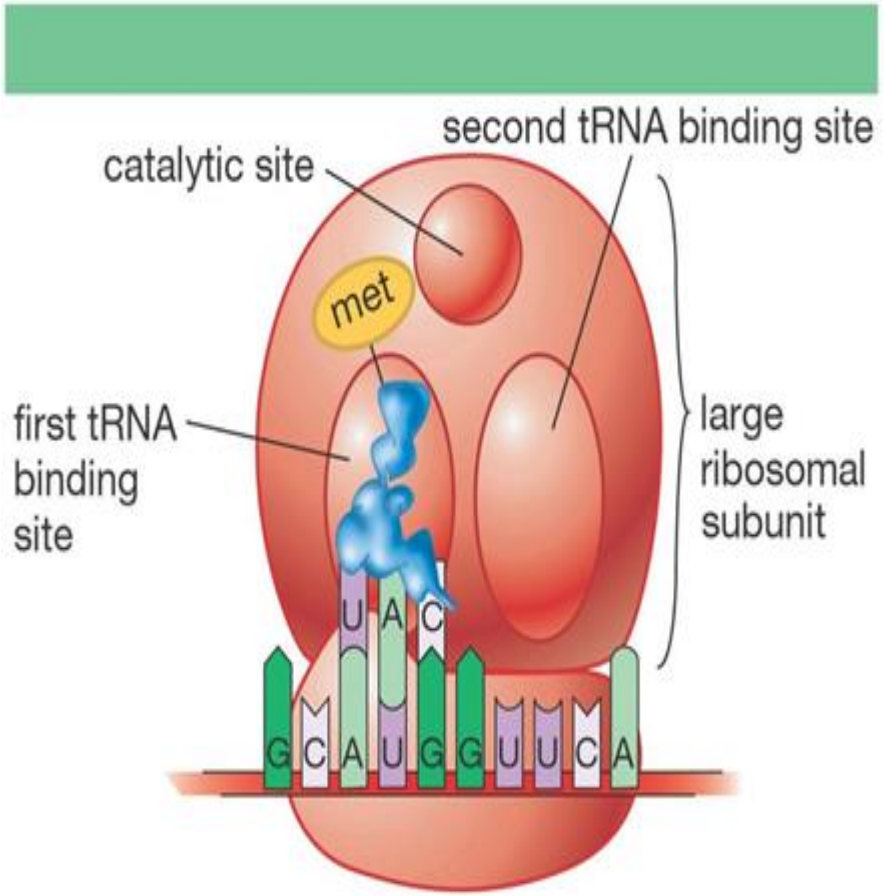


# The Genetic Code



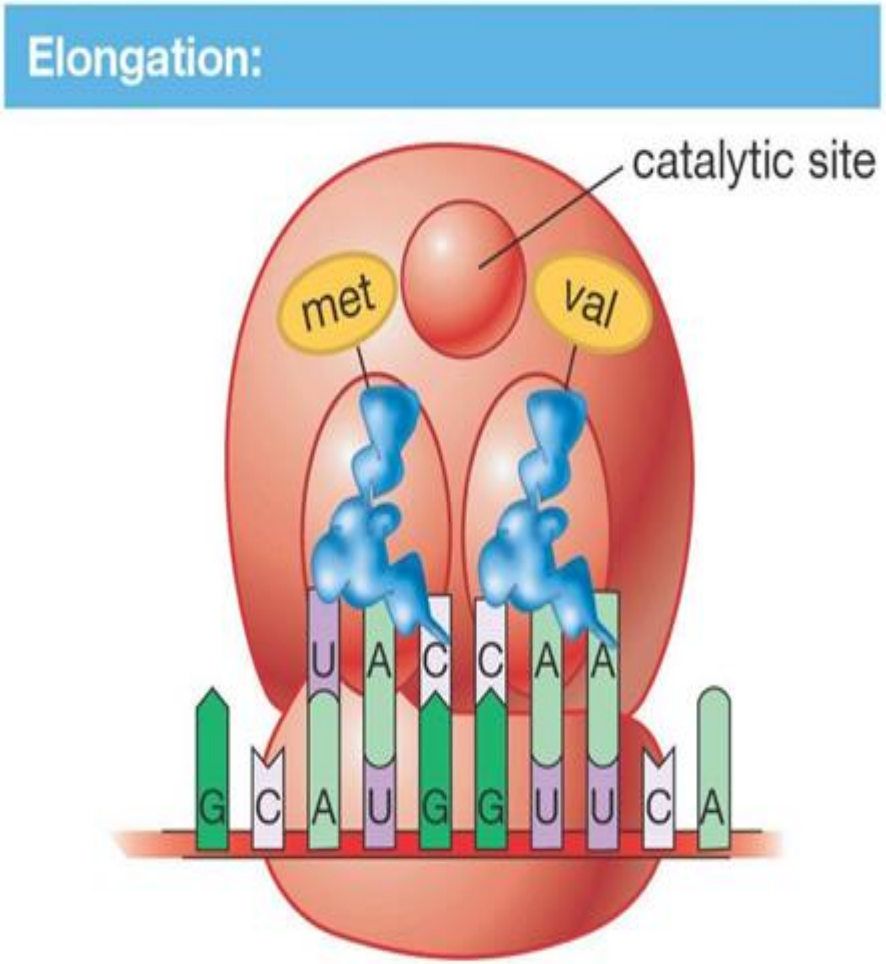
# Translation

The larger ribosomal subunit now binds to the smaller unit, forming a ribosomal complex. The tRNA binds to the first active site on the ribosome. Translation may now begin.



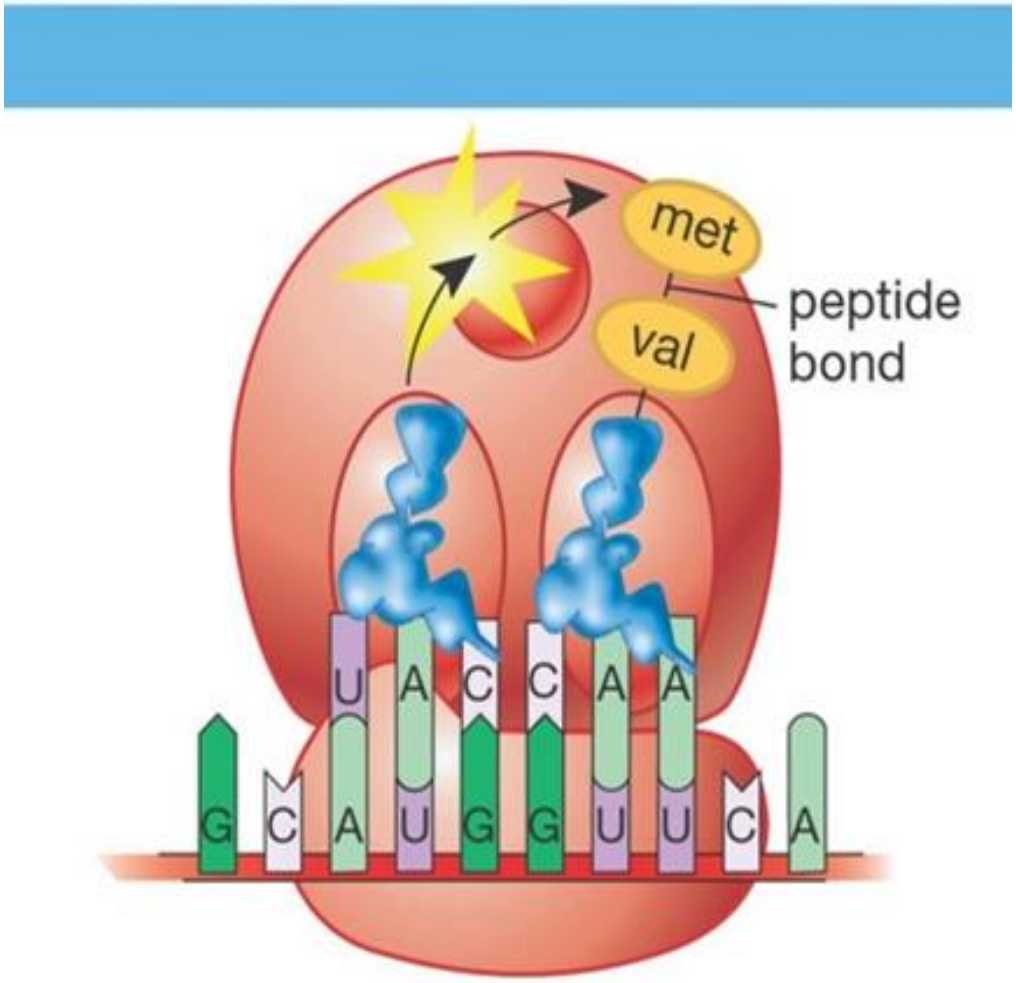
# Translation

The second codon in mRNA (GUU) matches the anticodon of a tRNA carrying the amino acid valine (CAA). The second tRNA binds to the second active site on the large subunit.



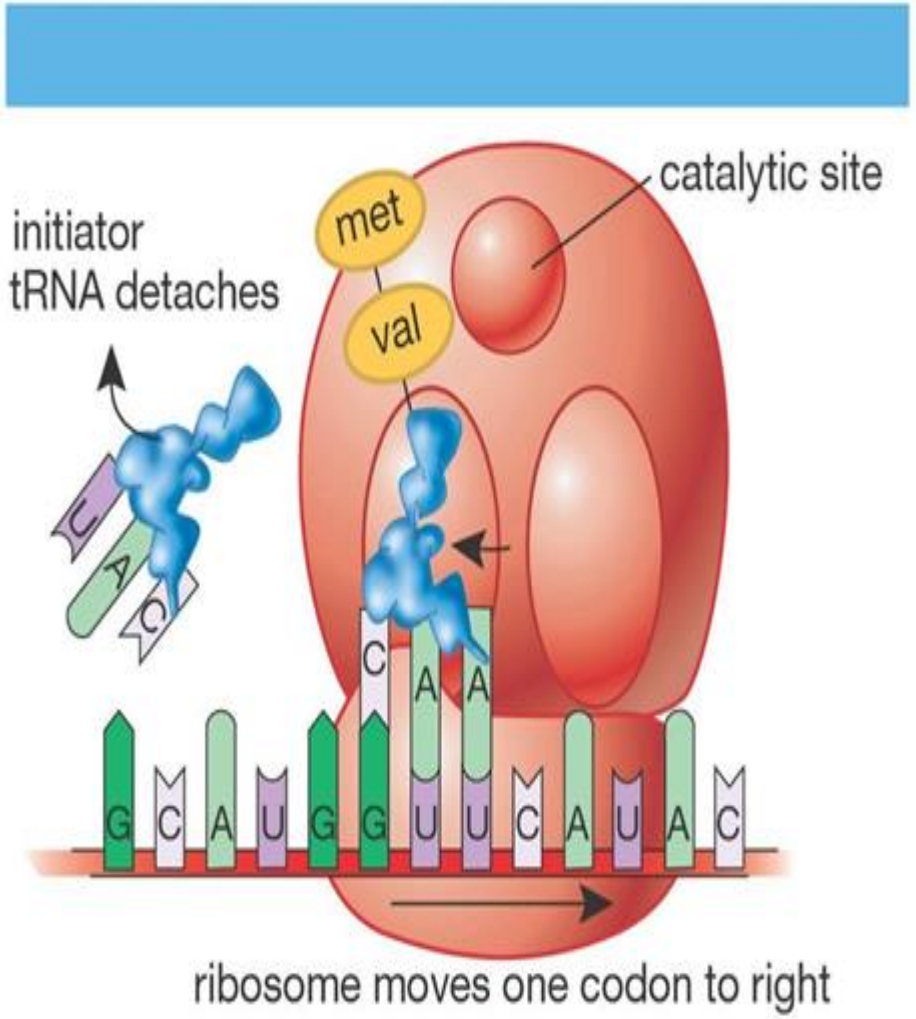
# Translation

A catalytic site on the larger subunit binds the two amino acids together using dehydration synthesis, forming a peptide bond between them.



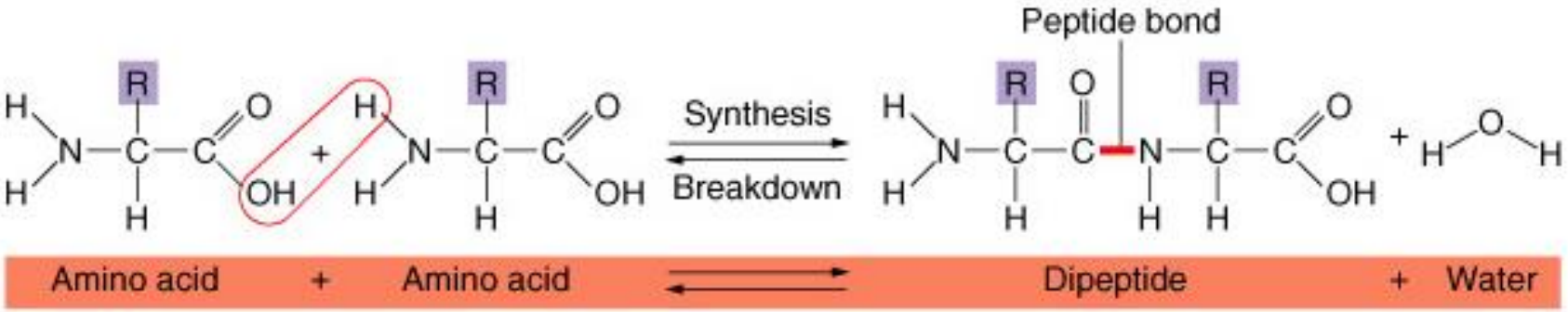
# Translation

The first tRNA now detaches and goes of to find another met in the cytoplasm. The mRNA chain shifts over one codon, placing the second codon (CAU) over the second active site.

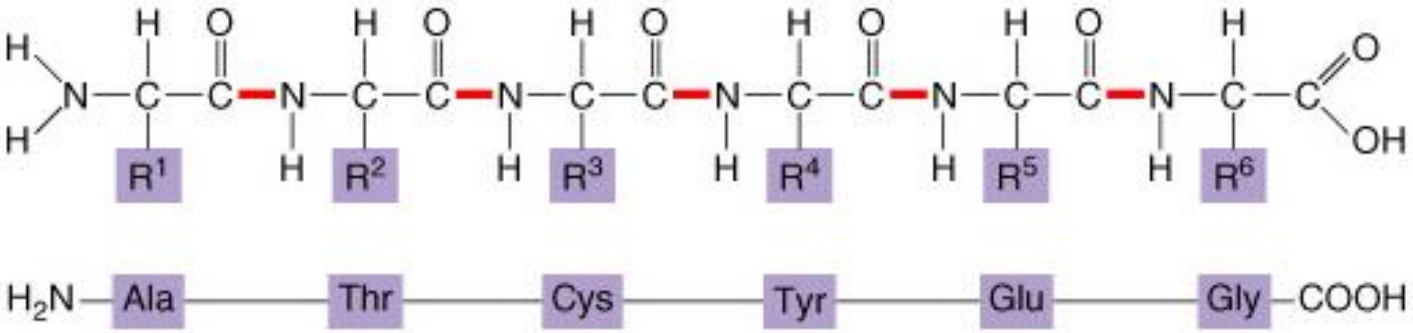


# Peptide Bonds

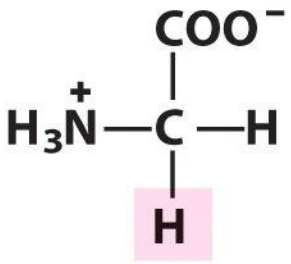
- Peptide bonds form between two amino acids:



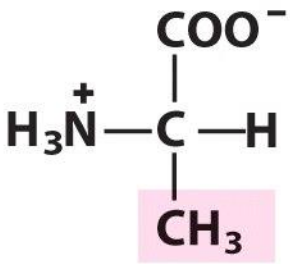

c.



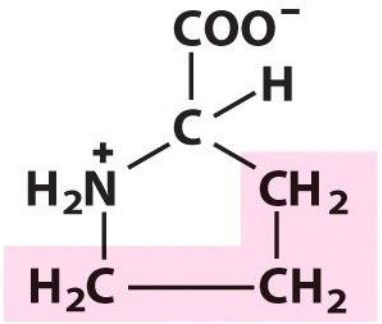
# Nonpolar, aliphatic R groups



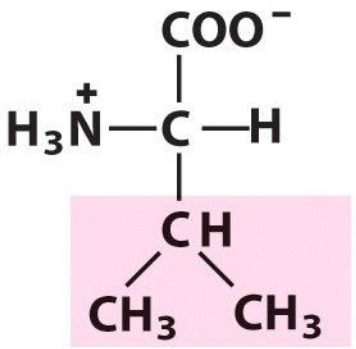
Glycine



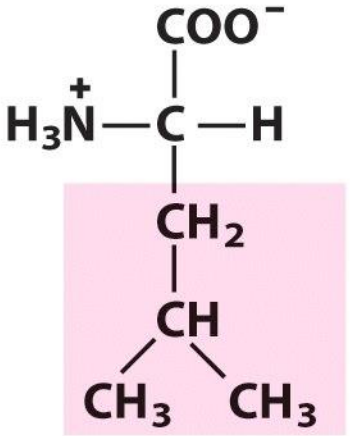
Alanine



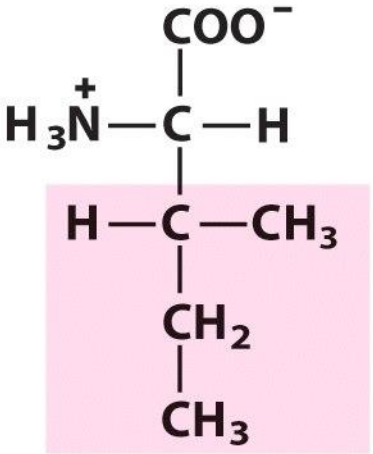
Proline



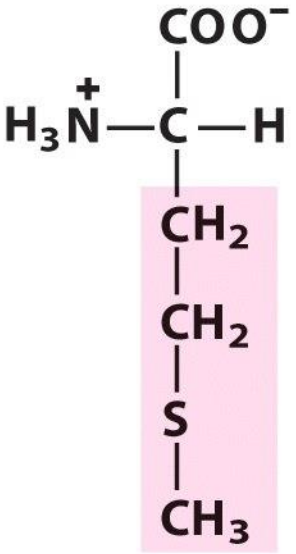
Valine



Leucine

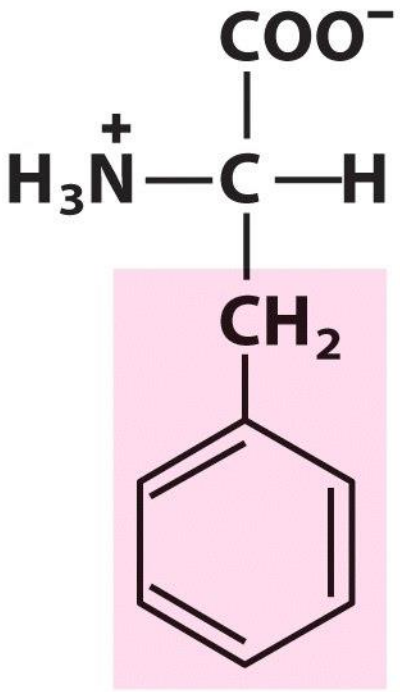


Isoleucine

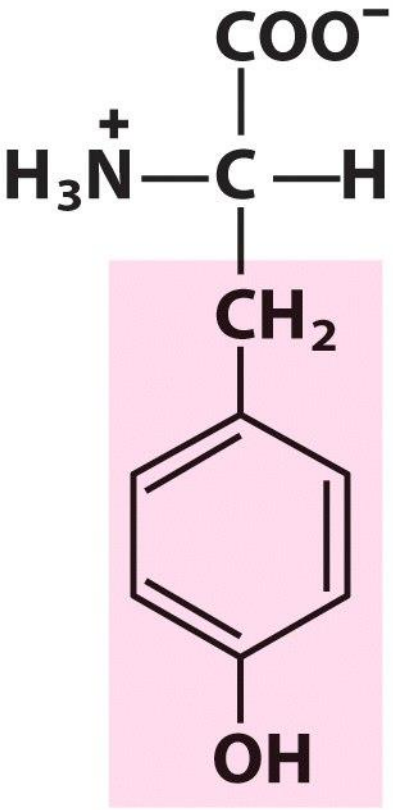


Methionine

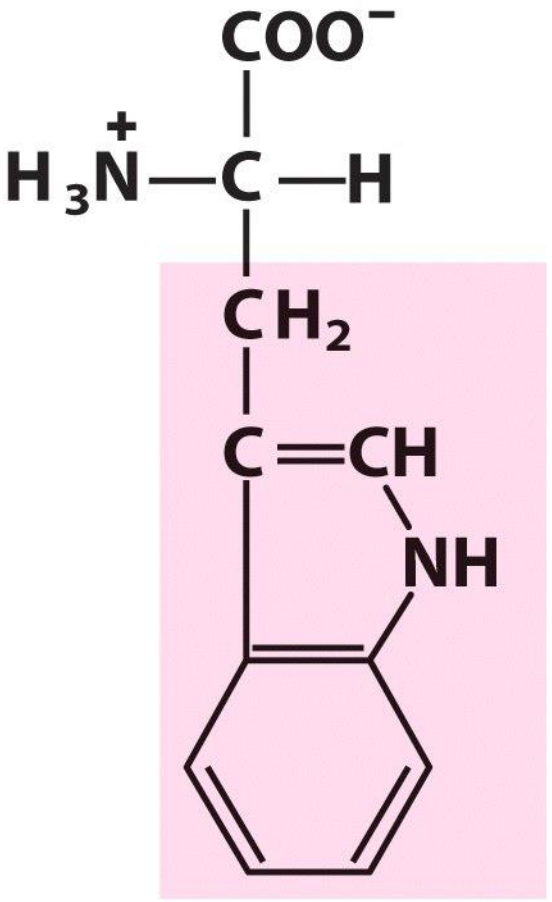
# Aromatic R groups




Phenylalanine

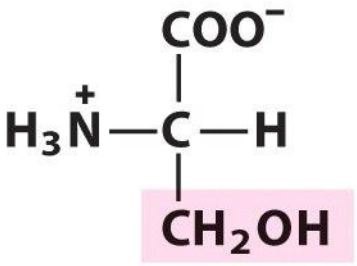



Tyrosine

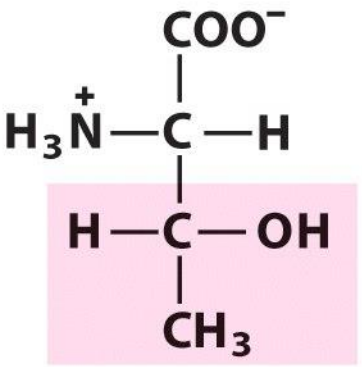



Tryptophan

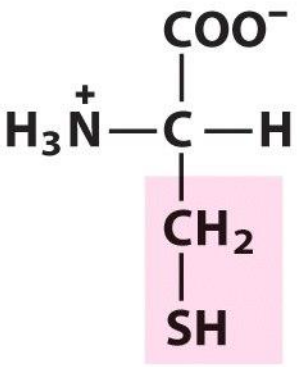
# Polar, uncharged R groups



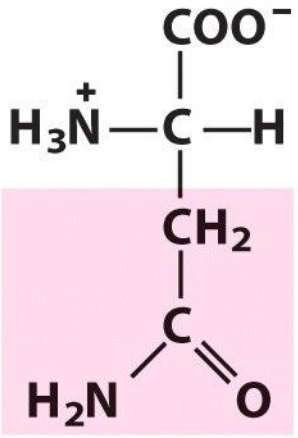

Serine



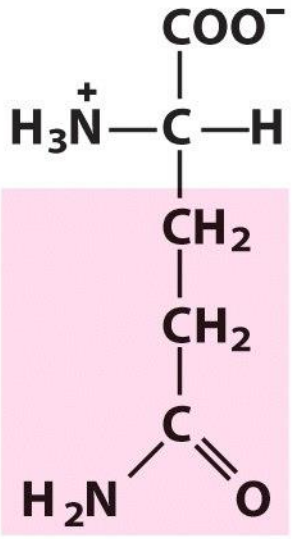

Threonine




Cysteine

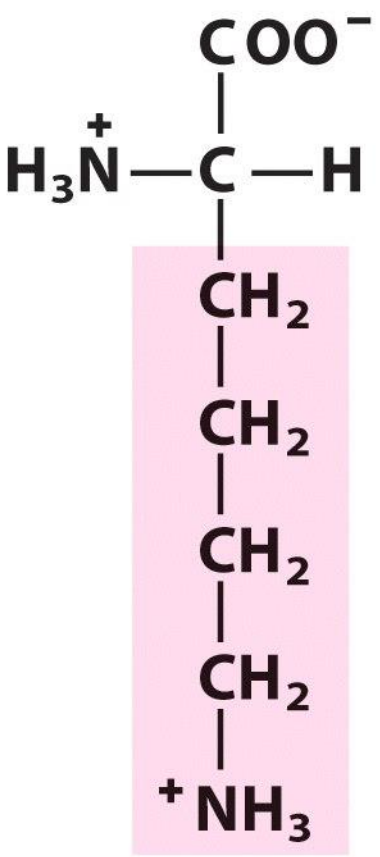



Asparagine

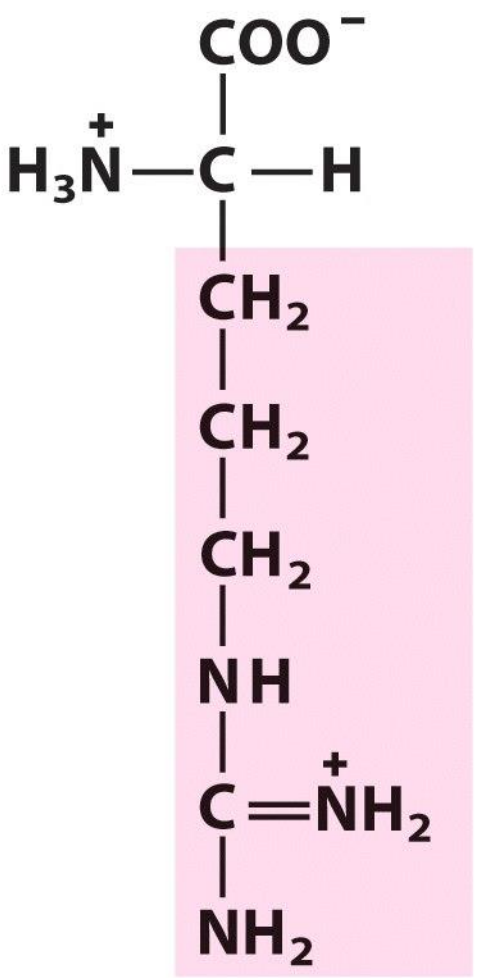



Glutamine

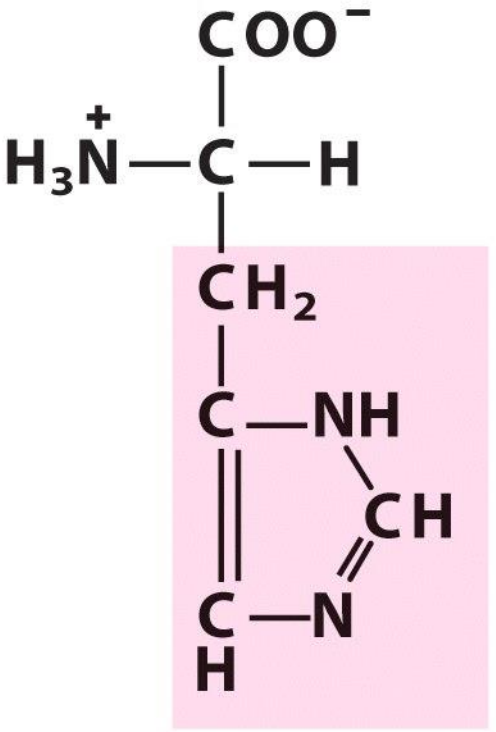
# Positively charged R groups



Lysine

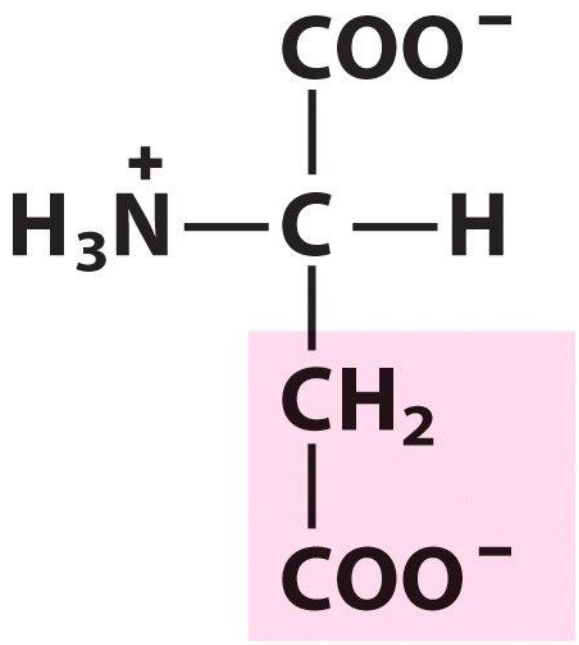


Arginine

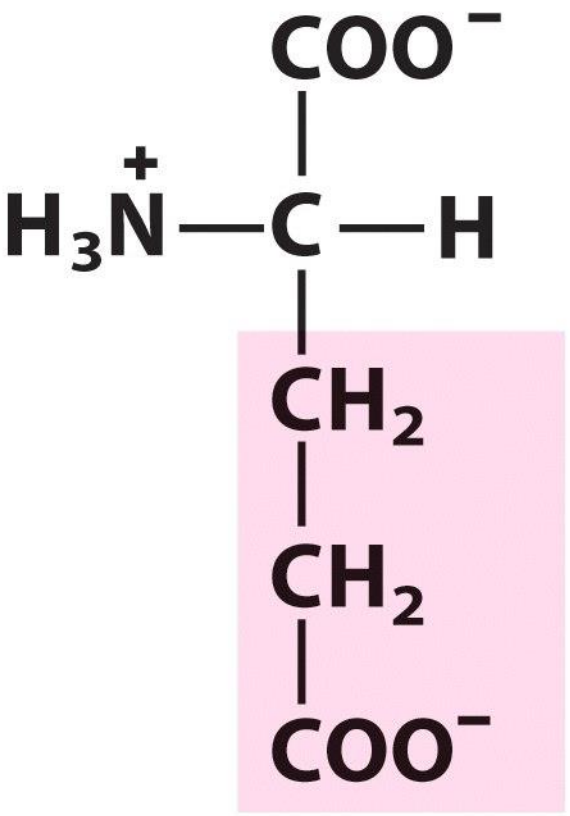


Histidine

# Negatively charged R groups



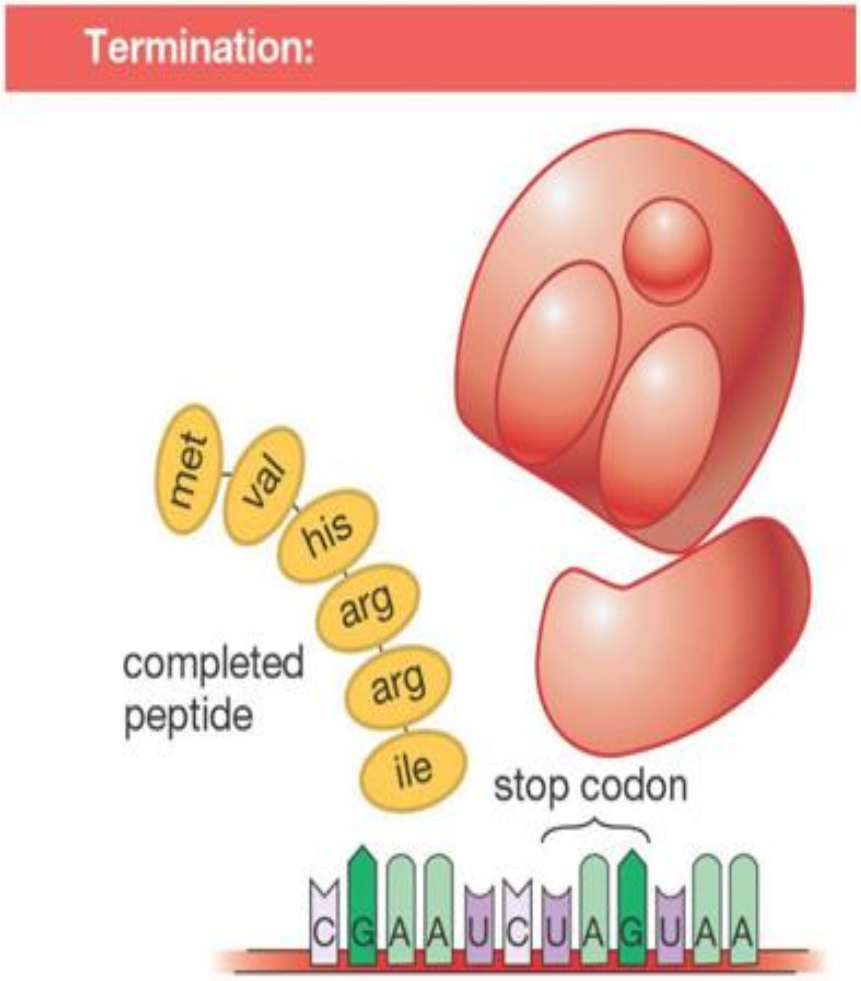

Aspartate




Glutamate

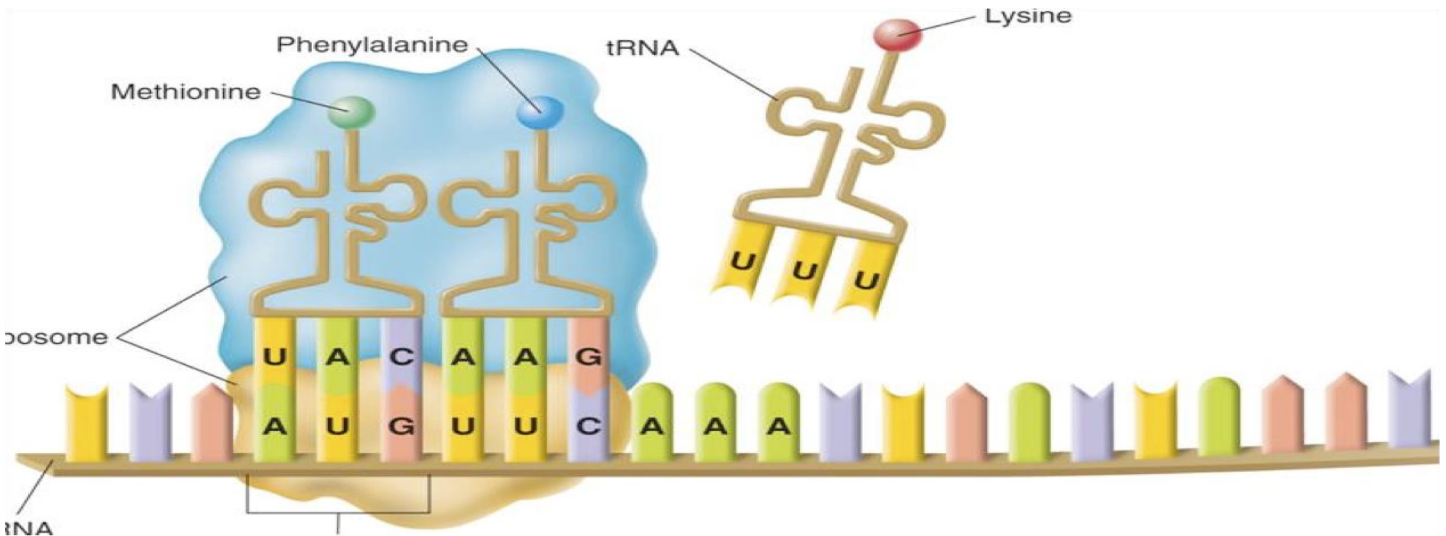
# Translation

The process continues until the ribosome finds a “stop” codon. The subunits detach from one another, the mRNA is released, and the polypeptide chain moves down the ER for further processing. The initial met is removed and the chain is folded into its final shape.



- **Codon**: 3-base code on mRNA that codes for a specific amino acid
  - Ex. CGU = alanine

    GUU = valine
- The sequence of bases in DNA tell mRNA what order **amino acids** must join together **to make a particular protein**.



# Watch clip



# Protein Folding

Proteins are the functional part of a cell

- Cellular “machinery” or tools

- Cannot function as a flat chain of amino acids



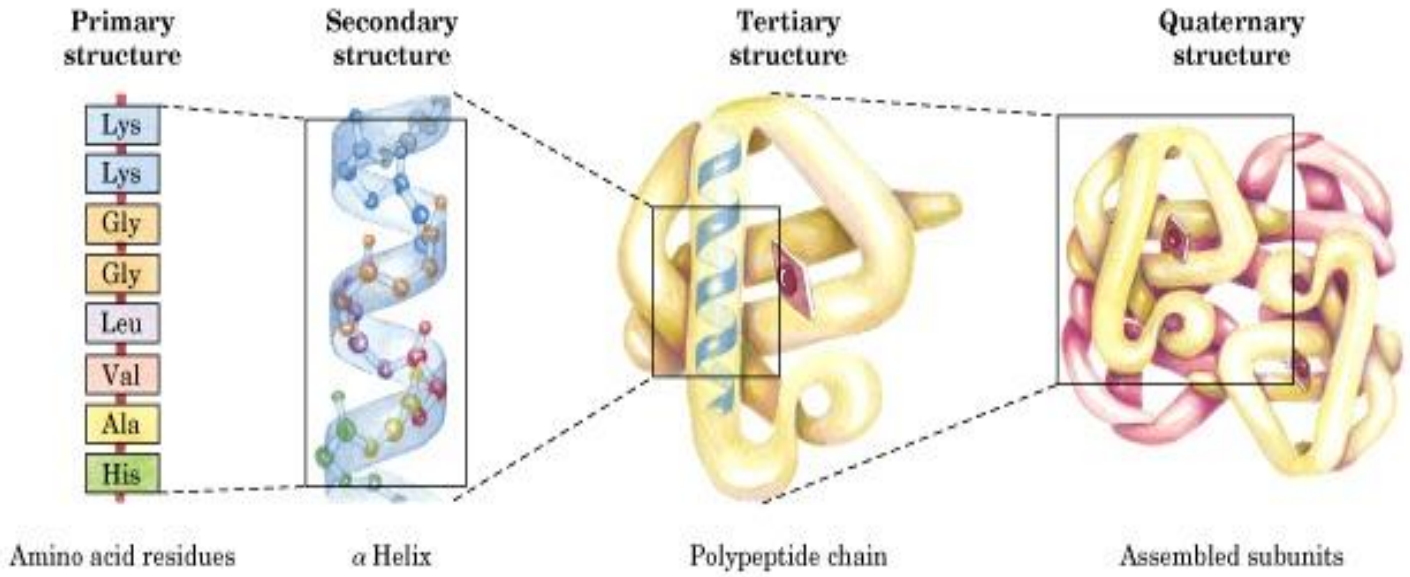
- Instead need to fold in very specific confirmations in order to function properly

# Protein's Structure

Four levels of structure:

1. Primary (1°) = amino acid sequence (order)
2. Secondary (2°) = loops, helixes, pleats, etc.
   - Caused by interactions between aa's
3. Tertiary (3°) = overall structure of entire protein
   - Caused by interactions with water
4. Quaternary (4°) = peptide subunits come together

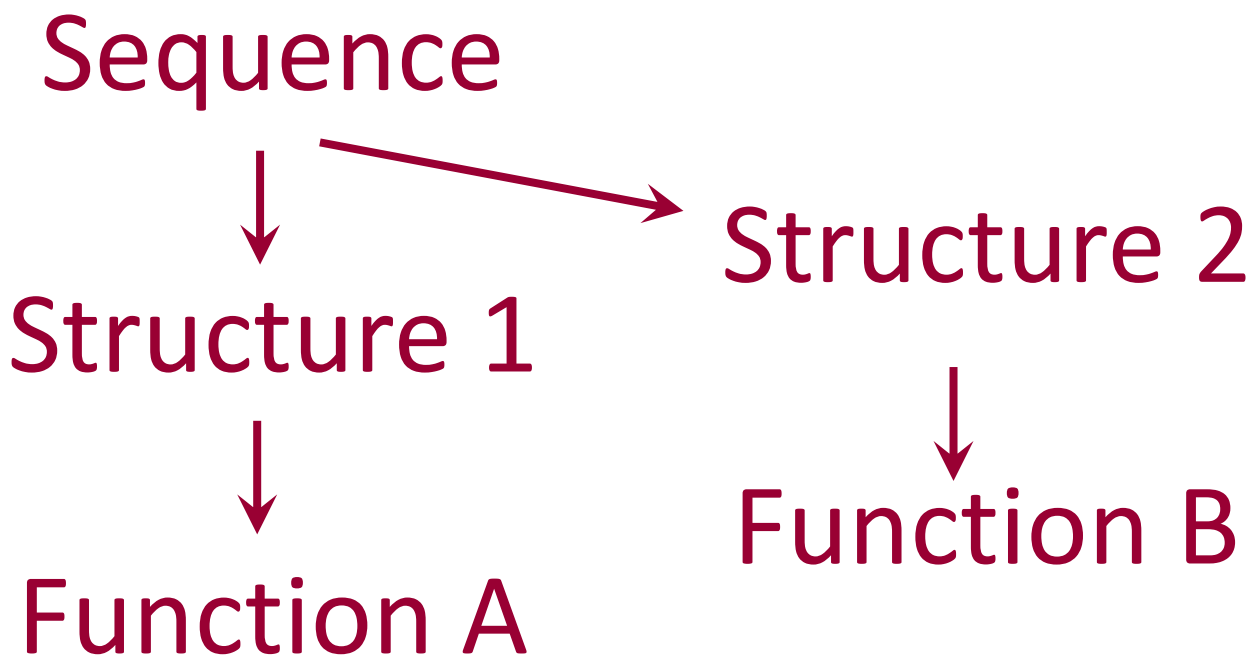
# The four levels of organization of protein structure



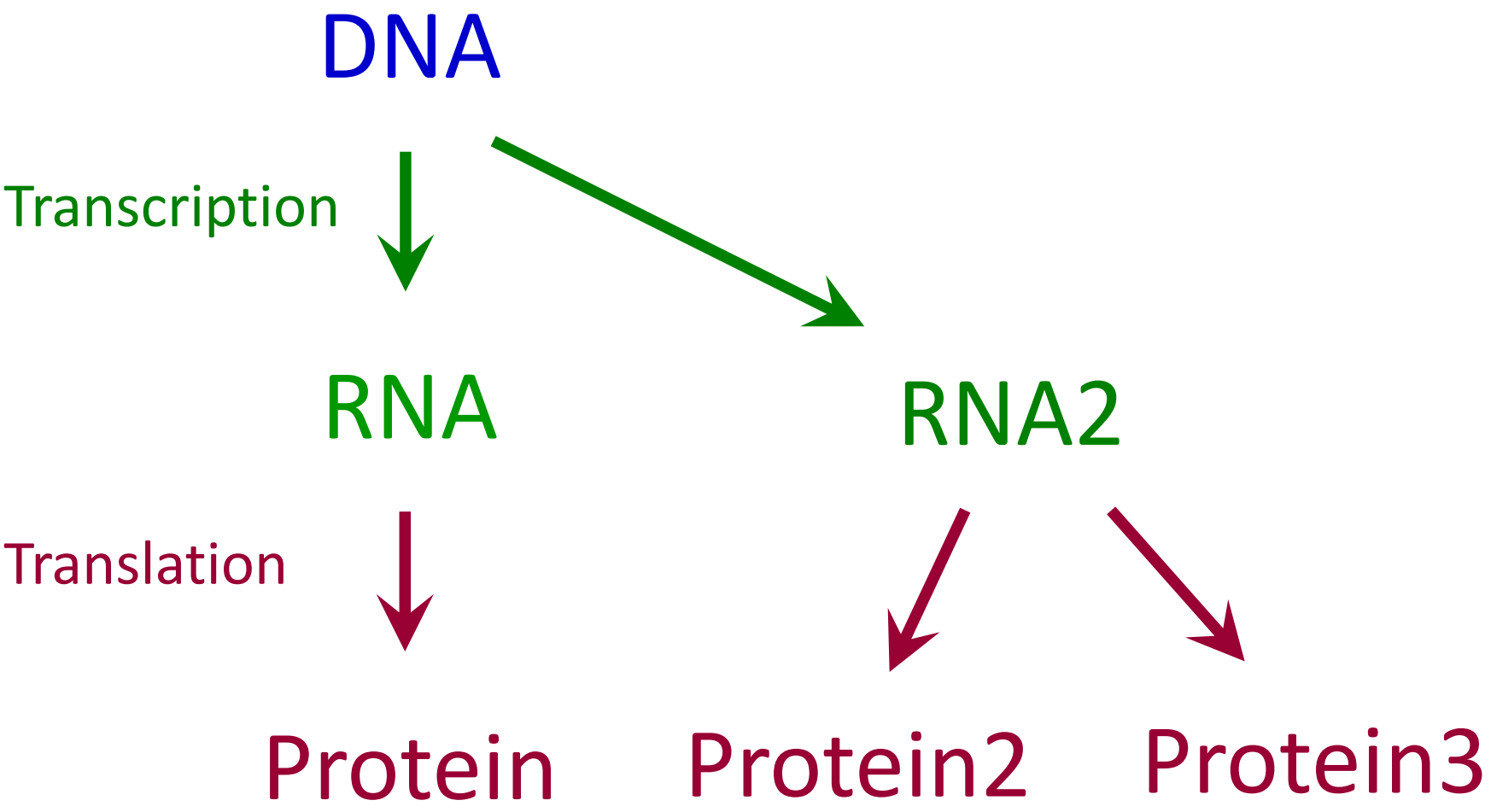

- **The primary structure of protein: a sequence of amino acids linked together by peptide bonds (covalent bond)**
- **The secondary structure of protein: Polypeptide folding into α helix, β sheet, or random coil (H bonds involved)**
- **The tertiary structure of protein: 3-D folding of a single polypeptide chain (H bonds, disulfide bonds, ionic bonds, van der Waals interactions, hydrophobic interactions)**
- **The quaternary structure of proteins: Association of two or more folded polypeptides (sub units) to form a multimeric protein (bonds and interactions similar to tertiary structure)**

# Proteins

- Sometimes protein sequence can fold in more than one way



# Central Dogma



Here is the sequence of a gene:

PROMOTER:

TTCCCTAGATAGAGATACTTTGCGCGCACACACATGCAAACGCGCGCAAAAAGG

AAAGCCCACCTATAAACTCCAGCCGCAAAGAGAAAACCGGAGCAGCCGCAGCTC

EXON 1:

ACCTGGCCGCGGGGCGGCGCGCTCGATCTACGCGTCCGGGGCCCCGCGGGGCCGG

GCCCGGAGTCGCCATGAATCGCTGCTGGGCGCTCTTCCTGTCTCTCTGCTGCTAC

CTGCGTCTGGTCAGCGCCGAG

INTRON 1:

GTGAGTTGCCACGGCGGCATGCAGTGGTTCGCCCCTTTTGGTGTCTGCCCGGCAG

EXON 2:

GGGACCCCATTCCCGAGGAGCTTTATGAGATGCTGAGTGACCACTCGATCCGCTCC

TTTGATGATCTCCAACGCCTGCTGCACGGAGACCCCGGA

INTRON 2:

GTAAATGGAATCCTCGCCCCGCGCTCCGGCCCTCCGAGGAGGTGGGG

CCGCCTGGTGTCTGACTGTGACTTCTCCTGCAG

EXON3:

AGGAAGATGGGGCCGAGTTGGACCTGAACATGACCCGCTCCCACTCTGGAGGCGA

GCTGGAGAGCTTGGCTCGTGGAAGAAGGAGCCTGG

Where does transcription start?

What is the sequence of the messenger RNA precursor and of the mature messenger RNA?

Where does translation start and what is the peptide sequence (first ten residues)?

Where does translation stop?